

## QA Lab Brasil

# Robot Framework

Open-source keyword-driven framework extensible through libraries for Web, API, desktop and other domains.

## What it is and when to use it

Robot Framework uses tabular syntax and keywords, supporting readable and reusable high-level automation.

## Prerequisites

Install Python 3 and pip. On Windows add Python to PATH. Verify `python --version` and `python -m pip --version`.

## Recommended virtual environment

Create a project folder, run `python -m venv .venv` and activate it. Use `.venv\Scripts\Activate.ps1` on PowerShell or `source .venv/bin/activate` on macOS/Linux.

## Step-by-step installation

Run `python -m pip install robotframework` and verify `robot --version`. Add a suitable Web library such as SeleniumLibrary or Browser Library when needed.

## First suite

Create a `.robot` file with Settings, Test Cases and Keywords and run `robot path/to/suite.robot`. Review `output.xml`, `log.html` and `report.html`.

## Keywords and organization

Design business-oriented keywords and keep resources, variables and libraries predictable.

## Libraries

Choose libraries according to the tested technology and project maturity.

## CI/CD and reports

Install requirements, run robot and publish log/report/evidence artifacts.

## Troubleshooting

Check PATH, active virtual environment, Python version and whether packages were installed into the correct interpreter.

## QA tip

Use the tool within a quality strategy: automate meaningful risks, keep the suite observable and treat automation failures as production-code issues.

## Official resources

Site oficial: <https://robotframework.org/>

Download / instalação: <https://robotframework.org/robotframework/latest/RobotFrameworkUserGuide>

Documentação: <https://docs.robotframework.org/>